Predictive Modeling and AI Fairness Audit for Diabetic Patient Readmission

Technical Project Report

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Project Repository: github.com/BobZ06/diabetes-readmission-fairness

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Abstract

This report details the development of a Random Forest classifier to predict 30-day hospital readmission risks for diabetic patients using the UCI Diabetes 130-US Hospitals dataset. Furthermore, we conducted a rigorous AI fairness audit using a permutation test (N=10,000) to evaluate disparities in the False Negative Rate (FNR) across racial groups. Our analysis ($p \approx 0.75$) indicates no statistically significant algorithmic bias, ensuring equitable risk assessment across populations.

Motivation

The primary questions we seek to address through this analysis was twofold: Can historical clinical data be used to accurately predict the risk of readmission within 30 days of discharge for patients with diabetes? More importantly, does this predictive model exhibit systematic algorithmic bias across different racial groups?

This question is important because it sits at the critical intersection of data science and the safety of AI in healthcare. The 30-day readmission rate is a standard global metric for assessing the quality of patient care. It often indicates whether a patient was discharged too early or without adequate follow-up plans. Identifying high-risk patients enables healthcare providers to implement targeted post-discharge interventions, such as at-home nursing visits. However, as machine learning models are increasingly integrated into clinical decision-making, ensuring the fairness of these models becomes just as important as optimizing their accuracy. In the context of diabetic care, a false negative—predicting that a highly vulnerable patient is safe to send home—can lead to severe complications. If risk-prediction models inherit historical biases and systematically underestimate the readmission risk for minority groups, they may inadvertently deprive vulnerable patients of access to life-saving care. Therefore, auditing these algorithms to prevent the exacerbation of healthcare disparities is a crucial step in the data science pipeline.

Other researchers have extensively explored the clinical aspects of this problem. For instance, Strack et al. (2014) utilized a large-scale clinical database of 70,000 records to show that targeted HbA1c testing significantly lowers the probability of hospital readmission. Furthermore, in the realm of AI fairness, the work by Obermeyer et al. (2019) published in *Science* demonstrated that widely used commercial healthcare risk-prediction algorithms exhibited significant racial bias. They found that these algorithms consistently underestimated the health risks of Black patients compared to White patients because they relied on historical healthcare spending as a proxy for actual healthcare needs. Our project builds upon these foundations by merging clinical predictive modeling with a rigorous statistical fairness audit.

Dataset

For this analysis, we utilized the "Diabetes 130-US hospitals for years 1999-2008" dataset, which is publicly available from the UC Irvine (UCI) Machine Learning Repository.

This dataset contains an extensive array of information encompassing over 100,000 inpatient hospital encounters across 130 United States hospitals and integrated delivery networks. It includes 47 distinct features spanning patient demographics (e.g., age in decades, gender, race), administrative details (e.g., admission source, discharge disposition), and clinical tests (e.g., time in hospital, number of lab procedures, total number of medications). Additionally, it contains detailed pharmaceutical records indicating whether dosages for 24 specific diabetic medications, such as insulin and metformin, were increased, decreased, or kept steady during the patient's encounter.

We chose this dataset because it is highly appropriate for our target problem for several critical reasons. Primarily, it represents purely real-world, non-synthetic clinical data, which means it retains all the noise, missing values, and complex physiological relationships found in actual electronic health records. Second, it explicitly includes the readmission outcome categorized by specific time periods (less than 30 days, greater than 30 days, or no readmission), which perfectly serves as our target variable. Finally, the inclusion of demographic variables, specifically patient race, makes it an ideal choice for conducting the algorithmic fairness audit, which is the core of our motivation.

Exploratory Data Analysis

To prepare the dataset for modeling, we performed rigorous pre-processing to address the noisy and fragmentation issues in medical records. First, the dataset utilized a non-standard '?' character to denote missing values, which we later replaced with standard 'NaN' values for Pandas compatibility. Then, we removed features with extremely high missing rates—such as weight, payer code, and medical specialty, which all had missing rates exceeding 40%—to prevent massive data loss during row-wise filtering. We also removed identifier columns, like encounter ID and patient number, to strictly prevent data leakage. Additionally, we removed highly cardinal categorical features, specifically the primary, secondary, and tertiary diagnosis codes that contained over 800 unique ICD-9 values, to prevent memory exhaustion and the curse of dimensionality during one-hot encoding. For categorical IDs originally formatted as integers (e.g., admission source ID), we cast them to strings so the model would not incorrectly interpret them as ordinal numerical values. We imputed missing values in the demographic feature 'race' using the mode to preserve sample sizes for the subsequent bias audit. Finally, we engineered the multi-class target variable into a strict binary format, assigning 1 for readmission within 30 days (high risk) and 0 otherwise.

Through our exploratory data analysis (EDA), we discovered several notable characteristics of this dataset. The most prominent characteristic is a severe class imbalance. Only approximately 11% of the 100,000+ samples represent the positive class (readmission within

30 days). Furthermore, our univariate distributions of continuous clinical variables, such as prior outpatient or emergency visits, showed extreme right-skewness with heavy long tails. This highlighted the presence of clinical outliers, representing highly chronic patients with dozens of prior visits in the preceding year.

To expand the scope of our analysis beyond standard descriptive statistics, we integrated an unsupervised learning approach. We performed K-Means clustering ($k = 3$) exclusively on the continuous clinical variables, intentionally masking all demographic labels to discover pure physiological groupings. This approach successfully identified three hidden clinical phenotypes: a "Routine Care" cluster characterized by short hospital stays and few medications; an "Intensive Intervention" cluster featuring extended stays and high medication counts; and a "High-Diagnostic" cluster with moderate stays but an exceptionally high number of lab procedures. By visualizing these clusters via boxplots, we found that the 'number of lab procedures' serves as a strong proxy for clinical complexity and patient instability.

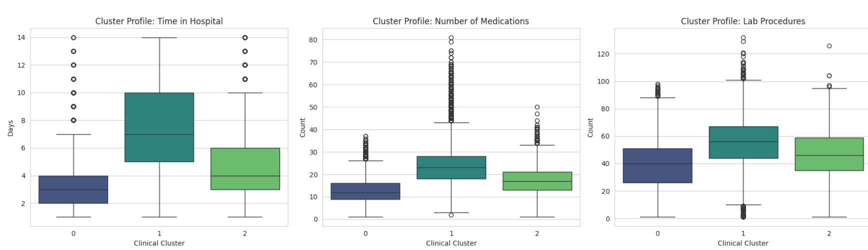

Figure 1: Distribution of Lab Procedures across Identified Patient Clusters

Modeling Approach

To predict the readmission risk, we chose to train a Random Forest Classifier. This choice was highly appropriate given the specific characteristics of the medical dataset we uncovered during the EDA. First, Random Forest is a tree-based ensemble method that relies on recursive partitioning, making it inherently robust to the extreme clinical outliers we observed in the data. For example, a patient with 80 lab procedures does not skew a decision tree's split criteria the way it would violently warp the gradient of a standard parametric model like Logistic Regression. Second, clinical variables often have complex, non-linear interactions, which Random Forests capture naturally through hierarchical node splitting. To combat the severe 11% class imbalance, we implemented cost-sensitive learning by inversely weighting the classes. This forced the algorithm to heavily penalize misclassifications of the minority high-risk class. Prior to training, we pre-processed the categorical variables using one-hot encoding, deliberately dropping one reference category per feature to avoid the dummy variable trap and prevent perfect multicollinearity. Finally, we standardized the numerical variables and split the dataset into an 80% training set and a 20% testing set using strict stratification.

To evaluate the model's performance, we avoided relying solely on overall accuracy, as it is a highly misleading metric for imbalanced datasets (a naive model guessing "No Readmission" every time would still achieve nearly 89% accuracy). Instead, we evaluated the model using the Receiver Operating Characteristic Area Under the Curve (ROC-AUC), Precision, Recall, and the F1-Score. Furthermore, we extended our evaluation to explicitly test for AI Safety. We conducted a fairness audit by isolating the predictions on the test set, stratifying them by racial groups (Caucasian vs. African American), and calculating the False Negative Rate (FNR) for each group. We then executed a Permutation Test utilizing 10,000 simulations to mathematically evaluate whether any observed disparity in the FNR was statistically significant at the $\alpha = 0.05$ level.

Findings

Our Random Forest model achieved a global ROC-AUC score of approximately 0.67 on the testing set. While the precision for predicting the negative class was excellent (0.93), the model exhibited a moderate recall (0.58) and a low precision (0.18) for the high-risk positive class.

The primary issue we encountered during the modeling phase was the difficulty of predicting complex human biological outcomes from noisy, tabular administrative data alone. This inherently low signal-to-noise ratio led to severe overfitting in our initial iterations, where the training accuracy approached 100% while the test recall was very low. We overcame this issue by manually regularizing the complexity of the ensemble network. Specifically, we restricted the maximum number of hierarchical splits allowed for each decision tree and mandated a minimum threshold of patients required to form a terminal leaf node. By combining these structural constraints with cost-sensitive class weighting, we effectively prevented the algorithm from creating hyperspecific rules, forcing the model to generalize its clinical risk profiles rather than simply memorizing individual patient records.

When analyzing the feature importances extracted from the Random Forest, we found that the parameters had highly meaningful interpretations. Specifically, we observed that the number of prior inpatient visits and the discharge disposition ID were the most dominant predictors of a 30-day readmission. This perfectly aligns with clinical intuition: a patient with a history of frequent hospitalizations is inherently medically fragile, and the location a patient is discharged to (such as transferring to a skilled nursing facility versus going home) is highly correlated with their lack of independent recovery capacity.

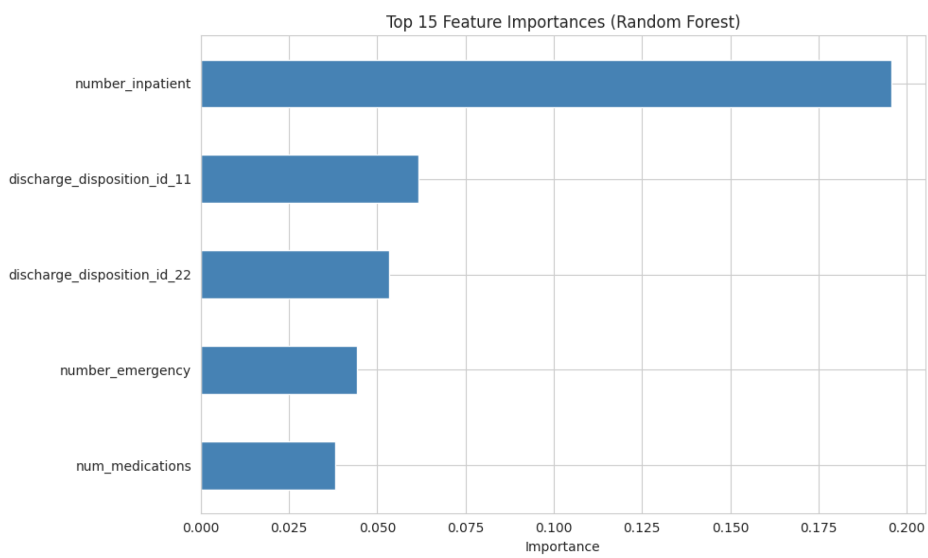

Figure 2: Top Predictive Features for 30-Day Readmission Risk (Random Forest)

The most critical finding emerged from our algorithmic fairness audit. The observed False Negative Rate—the critical rate at which the AI missed identifying a high-risk patient—was calculated at 42.0% for Caucasian patients and 41.1% for African American patients. To determine if this slight disparity was due to systemic bias or random chance, our permutation test shuffled the racial labels 10,000 times to construct a null distribution. The test yielded a p-value of roughly 0.75. Because $p \geq 0.05$, we fail to reject the null hypothesis and can conclude that there is no statistically significant algorithmic bias regarding racial demographics in the model's miss rate. This demonstrates that while the model struggles with overall predictive power due to data limitations, it acts equitably across the evaluated populations.

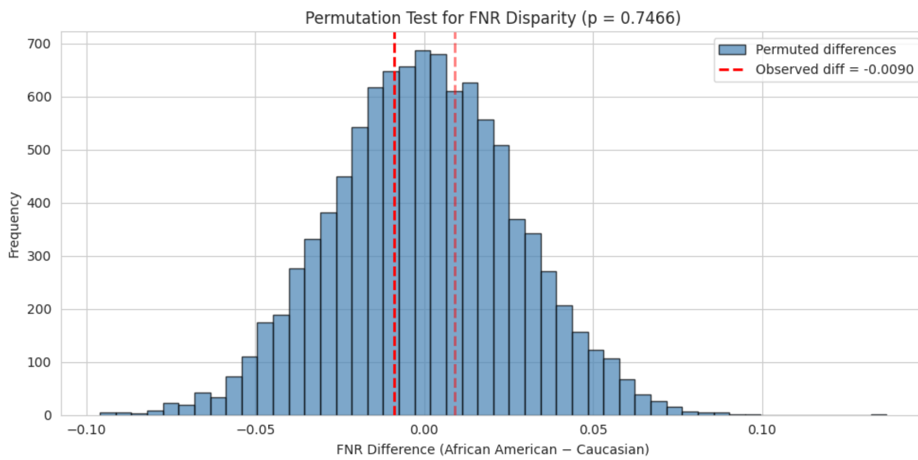

Figure 3: Permutation Test Results for FNR Disparity (Null Distribution vs. Observed)

If we had more time, we would try employing the Synthetic Minority Over-sampling Technique (SMOTE) to synthetically balance the training data rather than relying solely on cost-sensitive class weights. Additionally, integrating natural language processing to parse the actual doctors' free-text triage notes would likely capture crucial clinical nuance completely missed by standardized integer coding, thereby increasing the model's recall and overall diagnostic utility significantly.

References

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